Workflow for creating reproducible research articles using showyourwork

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Aim of DAPHNE4NFDI

Motivation

Ensuring reproducibility in scientific publications remains a major challenge [1], also for photon and neutron experiments. To address this, DAPHNE4NFDI aims to develop a bidirectional workflow that links raw data and publications via processing scripts used for data reduction and analysis scripts used for data analysis.

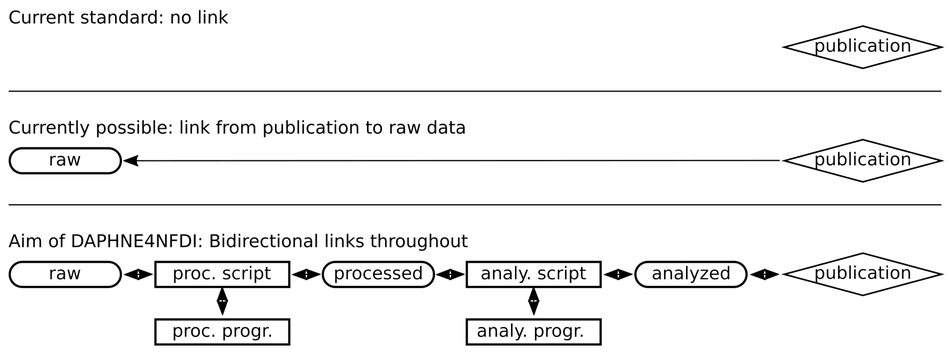

Figure 1: Snapshot from DAPHNE4NFDI proposal [2]: [Top] Current standard of publication without any raw data, [middle] currently possible situation of linking publication to raw data from facility, and [bottom] inclusion of processing (or reduction) and analysis steps between raw data and publication with bidirectional links.

Showyourwork philosophy for reproducing scientific articles

Philosphy

Showyourwork! [3] is an open-source reproducibility framework designed to enable transparent and fully reproducible scientific research. It organizes analysis scripts, software dependencies, and the manuscript within a GitHub repository, allowing any independent researcher to clone the project and regenerate the complete article. The manuscript preparation can be integrated with Overleaf, enabling collaborative writing and seamless synchronization between analysis and documentation.

Once reproduced, figures can be modified directly by editing the underlying scripts that generate them. Raw datasets are typically accessed from a Zenodo repository with read-only permissions, while intermediate and processed data products can be automatically uploaded to a separate Zenodo repository with readwrite access.

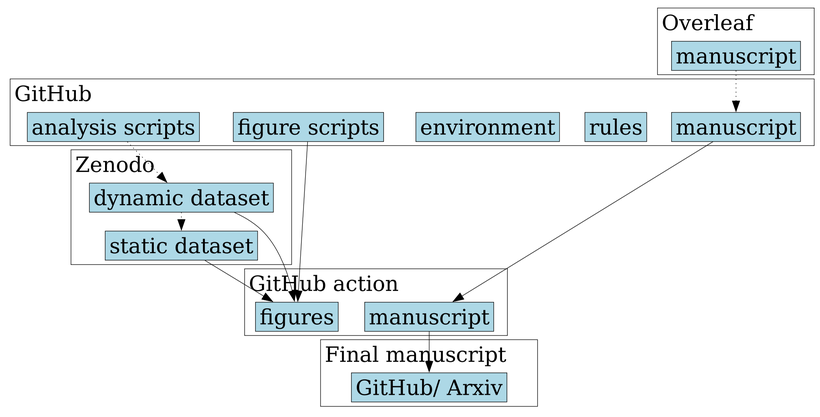

Figure 2: Framework of *showyourwork!*: GitHub hosts analysis scripts, environment files, dependency rules, and the manuscript; Zenodo stores datasets; and Overleaf enables manuscript editing. Commits trigger automated figure generation and manuscript compilation via GitHub Actions, producing a final version and an arXiv-ready tarball.

Demonstration of reproducible workflow using showyourwork

Showyourwork framework for this poster

This poster is created using the *showyourwork!* software, and an electronic copy is available under the Creative Commons Attribution 4.0 International license. The electronic copy can be downloaded from a GitHub repository [4], publicly available under the MIT license. Each figure caption in the electronic copy contains a maximum of three icons: The top two icons link to the dynamic and static Zenodo datasets used in the corresponding figure, and the bottom icon links to the script used to make the figure.

Application to a use case

A dummy use case is proposed in this work to demonstrate the essential features of the showyourwork and its relevance in scientific publications related to photon and neutron experiments. In the proposed use case, raw *measurement amplitude data* is collected using a 1D detector. The raw data, shown in figure 3, includes a background with linearly dependent on pixel coordinate. This background is removed to produce reduced data, as shown in figure 4. The reduced data is then fitted to $sin(kx+t)$ w.r.t. k and t shown in figure 5, where k is angular frequency, t is phase shift, and x is pixel positions.

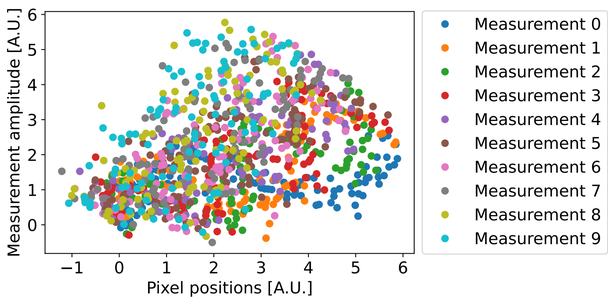

Figure 3: Raw data: Shown are amplitude vs pixel positions data with background for 10 different measurements. This plot is created by a GitHub-hosted plot script [4], which accesses the raw data from Zenodo [5]. The raw dataset and corresponding plot script can be directly accessed by clicking the dataset and GitHub icon on the left.

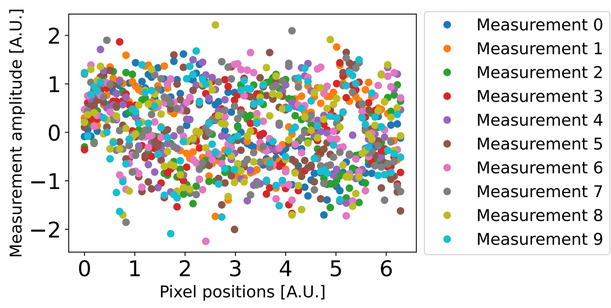

Figure 4: Reduced data: Shown are amplitude vs pixel positions data for all measurements after background removal from raw data. Within showyourwork, the raw data [5] is accessed and reduced by a script that stores the reduced dataset in Zenodo [6]. A plot script [4] accesses the reduced data and plots this figure. The reduced dataset (top icon), raw dataset (middle icon), and related plot script (bottom icon) can be accessed by clicking the icons shown on the left.

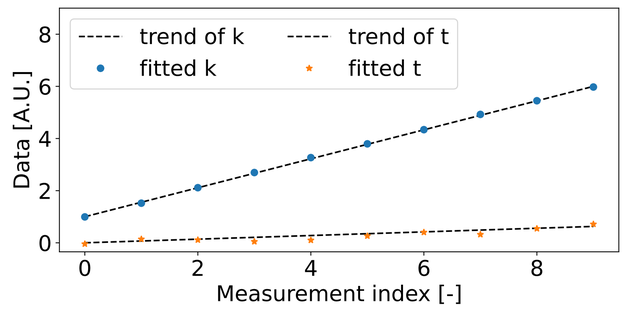

Figure 5: Analyzed data: Fitted values angular frequencies (k) and phase shifts (t) extracted from reduced data of different measurements are compared with the expected trend. Within showyourwork, the reduced data [6] is accessed and analyzed by a script that stores the reduced dataset in Zenodo [6]. The analyzed data (top icon), reduced dataset (top icon), raw dataset (middle icon), and related plot script (bottom icon) can be accessed by clicking the icons shown on the left.

Declaration

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References

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[4] Arnab Majumdar et al. Workflow for creating a reproducible research article using showyourwork, 03 2026. URL https://github.com/majumdararn/syw-poster-dam.

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